Energy

En analys från Swedish Subsidiarity

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Energy: the centralization paradox

Internal energy colonialism

Sweden has four electricity price areas (SE1–SE4) with dramatic price differences—from 20 öre/kWh in the north to over 300 öre/kWh in the south during the winter of 2022. At the same time, northern Sweden exports clean hydropower to the continent while southern Sweden imports dirty coal power from Germany and Poland. The base industry in Norrland is threatened by energy shortages at the same time as wind turbines are stopped by local protests and appeals.

The paradox: Sweden's electricity system is technically interconnected but politically and economically fragmented. Decisions on new production are made centrally (the state through Vattenfall, Svenska kraftnät) while the consequences are borne locally (closed reindeer grazing lands, disturbed views, pressured electricity prices). The result is "internal energy colonialism"—the north produces, the south consumes, and local communities have almost no influence.

The cybernetic analysis: The energy system requires both technical synchronization (low variation, central coordination) and local acceptance (high variation, local preferences). Today's model centralizes decisions on production and the grid, which creates a control unit with low variation that cannot match the local variation in landscapes, environmental interests, and businesses. The result: conflicts, appeals, and lost potential.

Active system failures (in brief)

- **The electricity price areas**: SE1–SE4 create enormous price differences without compensation mechanisms. Households in Norrland subsidize the south via grid fees, while the south suffers from undercapacity.
- **Vattenfall's monopoly position**: The state owns Sweden's largest power producer. No incentives for decentralized production or local ownership models. Vattenfall builds large-scale, not small-scale.
- **The permitting misery of wind power**: A wind farm requires permits from the county administrative board, the land and environment court, the Armed Forces, the Transport Agency, and often a municipal veto. The process takes 5–10 years.
- **The bureaucratic labyrinth of solar power**: Small-scale solar power (individuals, housing cooperatives) is hindered by complicated rules for grid connection, taxes on self-produced electricity, and disinterested grid owners.
- **The unfairness of grid fees**: Local electricity grid companies have monopolies and can charge high fees. The customer cannot choose—a statutory monopoly lacking local influence.

- **Lack of energy storage and flexibility**: Centralized planning has underestimated the need for batteries, hydrogen, and demand flexibility. Sweden is 5–10 years behind other countries.

Subsidiarity-based measures

Here are five action areas. Each area has its own page with concrete pilots, budgets, rule changes, and analysis of resistance.

Measure	Short description
→ Energy cooperatives	Local energy cooperatives that own and operate wind power, solar parks, or hydropower. Members share the surplus and get priority for green electricity. Model from Denmark (Samsø) and Germany.
→ Peer-to-peer electricity trading	Enables direct trading of solar electricity between neighbors (peer-to-peer) via a digital platform. Existing grids are used, but the customer can choose the seller. Requires an amendment to the Electricity Act.
→ Energy banks	Municipal "energy banks" where households and companies can lend surplus capacity (batteries, electric cars, heated water tanks) to the grid during peak loads. Prevents capacity shortages.
→ Municipal energy councils	Each municipality establishes an energy council with residents, business, and local grid owners. The council has veto power over new large-scale projects (wind power, power lines) that lack local acceptance.
→ Local grid fees	The electricity grid fee is determined locally by a user-cooperative board (not by the grid monopolist). Model from "municipal utility districts" in the USA.

Implementation: from theory to practice

Phase 1 (1–3 years) – Five municipalities (e.g., Piteå, Öland, Varberg, Borås, Åre) test energy cooperatives with a state start-up loan. A digital platform for peer-to-peer trading is piloted in a city district (e.g., Hammarby Sjöstad, Stockholm). Energy banks are installed in three local electricity grids.

Phase 2 (3–7 years) – Municipal energy councils gain formal veto power via an amendment to the Environmental Code. Legislative changes enable peer-to-peer trading and differentiated grid fees. Five grid companies are transformed into cooperatives.

Phase 3 (7–15 years) – Sweden reaches 100% renewables through a combination of central baseload (hydropower) and distributed flexibility (solar, wind, batteries, peer-to-peer trading). Local energy systems

are interconnected but independent.

Expected resistance and answers

Concern	Response
"Vattenfall is state-owned—we cannot compete with ourselves"	Vattenfall must be technology-neutral and market-oriented. Cooperative competition forces Vattenfall to become more efficient.
"Grid owners lose their monopoly—they will sabotage"	The grid monopoly is already inefficient. Cooperative grids in Germany and the USA have lower fees and higher customer satisfaction.
"Peer-to-peer trading undermines grid stability"	The technology exists (blockchain, smart meters). The grid owner's role remains—peer-to-peer trading uses the same physical grid.
"Municipal veto stops all wind power"	Municipalities already have veto power under the Planning and Building Act, but rarely use it. Formal veto power with clear criteria increases predictability.

The Swedish advantage

Sweden has **high trust** (enables cooperatives), **digital infrastructure** (smart electricity meters in all homes, BankID), **clean hydropower** (baseload), **municipal self-government** (enables local energy councils), and a **technically competent population**.

Conclusion: choose between centralized conflicts and distributed collaboration

Today's energy system is not technically immature—it is **architecturally flawed**. Decisions are made at the wrong level: production centrally, consequences locally. The result is conflicts, appeals, and lost acceptance for the green transition.

The choice is simple:

- Continue as today: More large-scale projects without local acceptance, longer lead times, and remaining price differences between north and south.
- Begin the transition: Energy cooperatives, peer-to-peer trading, energy banks, municipal energy councils, local grid fees.

Three things you can do tomorrow

If you are a municipal politician: Start an energy council—invite local companies, grid owners, and citizens. Map local resources (solar, wind, water, waste heat). Develop an energy plan that prioritizes local ownership.

If you are a homeowner: Join a solar cell network (e.g., on Facebook or a local energy cooperative). Buy solar cells together with neighbors—lower price per unit. Start talking to your grid owner about selling surplus directly to neighbors.

If you are a tenant or condominium owner: Submit a motion to your housing association to install solar cells on the roof, battery storage in the basement, and to sign agreements on peer-to-peer trading with neighboring properties.

This analysis applies cybernetic principles (Ashby's law of requisite variety) to Swedish energy systems. For deeper measures, click on the links above.

Energy cooperatives

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Summary: Local energy cooperatives that own and operate renewable electricity production—wind power, solar parks, small-scale hydropower, or biogas. The members (households, companies, municipalities) are co-owners, receive a share of the surplus, and have priority for the produced electricity at cost price. The model is taken from Denmark (Samsø, Middelgrunden) and Germany (over 1,000 energy cooperatives). The goal: to break Vattenfall's monopoly, increase local acceptance of green energy, and let those who bear the consequences of energy production also share the profits. Financing through green bonds, state start-up loans, and member contributions.

The mechanism

Today, Swedish electricity production is largely owned by the state (Vattenfall), large power companies (Uniper, Eon), and venture capital-owned wind power companies. Local communities see the wind turbines, power lines, and transformer stations—but receive no share of the revenue. The result: protests, appeals, and a drawn-out permitting process that delays the green transition by several years.

The measure: An **energy cooperative** is an economic association (cooperative) that:

- Is formed by at least 10 members—households, companies, organizations, or the municipality.
- Develops, finances, and operates one or more facilities for renewable electricity production (solar, wind, water, biogas).
- The members subscribe to shares (share capital) of, for example, SEK 10,000–50,000. The return comes from the sale of electricity (at market price or directly to the members).
- The cooperative can sell electricity on the open market (Nord Pool) or sign agreements directly with the members ("electricity price guaranteed at cost").
- The surplus (profit) is distributed to the members as interest on the share capital, or reinvested in new projects.

Advantages compared to large-scale commercial projects:

Aspect	Commercial company	Energy cooperative
Owner	Venture capital, large corporations	Local members (households, companies)

Aspect	Commercial company	Energy cooperative
Profit motive	Maximize profit to external owners	Back to members, often lower return requirements
Local acceptance	Low (external actors)	High (members are neighbors)
Permitting process	5–10 years, many appeals	2–4 years (if local acceptance exists)
Cost of capital	6–8% (venture capital)	3–4% (green bonds, member contributions, state loans)
Local economy	Money leaves the district	Money stays locally (interest, jobs, services)

Practical example (hypothetical Swedish case):

A group of 50 households in a village in Västerbotten forms a cooperative. They join forces with the local community association and a small sawmill. They plan four wind turbines (total capacity 12 MW) on a nearby ridge. The cooperative raises member contributions totaling SEK 5 million, a state start-up loan of SEK 10 million (interest-free for 3 years), and a green bond loan of SEK 80 million from Kommuninvest or the European Investment Bank. The facility generates an annual turnover of approx. SEK 15 million (with normal wind). After interest and operations, the profit is SEK 4 million per year—which is divided between the members (approx. SEK 80,000 per household per year) and reinvestments. The district also gets several local jobs (operations, maintenance, administration).

Pilot design – 10 cooperatives first

Pilot phase 1 (years 1-5): Ten energy cooperatives in different parts of the country receive state start-up support and advice. The selection is made via an application round to the Swedish Energy Agency (Energimyndigheten).

Proposed candidates (realistic based on geography and local engagement):

Location	Type	Members	Facility	Estimated capacity
Piteå	Solar park + battery	100 households, the municipality	5 hectares of land	3 MW solar + 2 MWh battery
Öland	Wind power	150 households, farms	4 turbines (2 MW each)	8 MW
Varberg	Offshore wind	200 households, local companies	6 turbines (3 MW)	18 MW (pilot)

Location	Type	Members	Facility	Estimated capacity
Borås	Biogas (farm-based)	50 farmers	Digestion plant	5 MW heat + electricity
Åre	Small-scale hydropower	50 households, tourism companies	2 existing mills renovated	1 MW
Skåne (Tomelilla)	Solar cell association	300 homeowners	15 hectares of arable land	10 MW solar
Dalarna (Leksand)	Mixed (solar+biogas)	Community association, municipality	Solar cells on roofs, farm biogas	2 MW electricity + heat
Västerbotten (Lycksele)	Wind power	Sami, farmers, villagers	3 turbines (3.6 MW)	10.8 MW
Gotland	Solar + battery storage	500 households (tourism industry)	20 hectares of land, battery park	15 MW solar + 10 MWh battery
Närke (Kumla)	Bioenergy (wood chips)	Several farms, district heating company	Wood chip boiler, ORC turbine	5 MW electricity, 20 MW heat

Steps to form a cooperative in the pilot:

1. **Interest group**: At least 10 people/companies sign a letter of intent.
2. **Feasibility study**: The cooperative receives a grant of SEK 100,000 (from the Swedish Energy Agency) to investigate technical, economic, and legal prerequisites.
3. **Capital raising**: The cooperative subscribes shares among its members. The state offers co-financing (start-up loan) up to 50% of the project cost, max SEK 10 million per cooperative. Repayment with interest (1–2%) after 5 years.
4. **Permits**: The cooperative applies for grid connection, environmental permits, and building permits. Pilot cooperatives receive priority in processing (max 12 months) and exemptions from certain requirements (e.g., conducting a full Environmental Impact Assessment if the capacity is below a certain level).
5. **Construction**: The cooperative procures contracting (can be local). Completion within 2 years.
6. **Operations**: Electricity is sold on Nord Pool or directly to members via agreements. The surplus is distributed.

Evaluation:

- Number of members per cooperative (target: >100)
- Average return on member contribution (target: 4–6% per year)

- Time from idea to operation (target: <4 years, compared to 8–10 years for commercial wind farms)
- Local acceptance (survey to residents in the area before/after)
- Change in the number of appeals (expected close to zero)

Budget and financing

Item	Amount (per pilot cooperative)	Total 10 cooperatives
Feasibility study grant (SEK 100,000)	SEK 0.1 million	SEK 1 million
State start-up loan (max SEK 10 million, repaid)	SEK 10 million on average	SEK 100 million (loan)
Interest subsidy (1% for 3 years)	SEK 0.3 million (per loan)	SEK 3 million (cost)
Advice (Swedish Energy Agency, cooperative center)	SEK 200,000	SEK 2 million
Evaluation	SEK 100,000	SEK 1 million
Total state cost (excl. loans to be repaid)	**SEK 0.7 million per cooperative**	**SEK 7 million**

Financing: SEK 7 million in direct costs (grants, interest subsidies, advice) is a very small investment compared to the socio-economic benefit. If each cooperative builds facilities for SEK 50–100 million (private and borrowed funds), an investment volume of SEK 500–1,000 million is created with minimal state input. Furthermore, appeals and lead times are reduced, which speeds up the green transition.

Regulatory changes required

1. **Amendment to the Electricity Act (1997:857)** – make it easier for small-scale producers to sell electricity directly to members ("direct agreements"). Today, the cooperative is required to register as an electricity trading company (heavy and expensive). Introduce an exemption for cooperatives under 50 GWh/year.
2. **Amendment to the Environmental Code (1998:808)** – introduce a simplified permitting process for renewable electricity production owned by local cooperatives (e.g., abolished EIA for wind power under 10 MW, if the cooperative and nearby residents are members). Consultation requirements remain, but the right to appeal is limited to members.
3. **Ordinance on state start-up loans** – the Swedish Energy Agency is tasked with establishing a loan guarantee for energy cooperatives, with 1–2% interest and an interest-only period for 3 years.
4. **Tax relief** – cooperatives that produce electricity for direct consumption among their members are exempted from energy tax (just as private solar electricity is already tax-free). Reduce administration.

Expected resistance and answers

"Vattenfall and the large power companies will lobby against it—they don't want competition."

Answer: Vattenfall is owned by the Swedish state. The state can instruct Vattenfall to support rather than oppose cooperatives, for example by offering technical services (grid connection, balance responsibility) at a reasonable cost. Cooperatives constitute a very small share of the market (max 5–10% in the long term)—the large companies will not lose noticeably, but they can gain from higher acceptance of wind power (which reduces the risk of future halts).

"This will just be for rich homeowners—those who have capital to invest."

Answer: Cooperatives can be designed with low contributions (e.g., SEK 1,000 per share) and member loans (e.g., tenants can join without a contribution, paying a higher electricity fee instead). The municipality can be a member and subsidize shares for low-income households. Experiences from Germany show that energy cooperatives recruit broadly, including pensioners and young people. The pilot budget includes funds for advice that can be directed toward broadened participation.

"The permitting process is already long—an exemption for cooperatives creates unequal treatment."

Answer: The point is exactly to correct a distortion. Commercial companies can handle long processes (they have capital and lawyers). Cooperatives do not have the same resources. A simplified procedure for local, member-owned projects is proportionate and increases the net amount of green energy—because otherwise they would not be built at all.

"Won't the grid owners (Eon, Vattenfall Eldistribution) hinder connection?"

Answer: The grid owner is obliged to connect all approved production facilities according to the Electricity Act. Cooperatives can appeal unjustified obstacles to the Swedish Energy Markets Inspectorate (Ei). The pilot's legal support includes help with such disputes. Experiences from other countries (Germany, Denmark) show that grid owners cooperate when there are political directives.

"Are there any Swedish examples?"

Answer: Yes, some small-scale ones. **Kraftringen** in Lund (municipally owned, not cooperative). **The Oxhagsverket wind power cooperative** in Skåne (formed by villagers, around since the 1990s). **Southern Sweden's Energy Cooperative** (smaller). But no major spread. Denmark has over 100,000 households in

wind cooperatives. Germany has 1,000+ energy cooperatives. Sweden is lagging behind—the pilot can start a popular movement.

Swedish examples to build upon

- **Oxhagsverket (Skåne)** – Sweden's oldest wind cooperative, built in 1991, still in operation. Members are local farmers and households. The model works.
- **Lotsbroverket (Värmdö)** – smaller cooperatively owned wind turbine. Inspired by Denmark.
- **Bygdealiansen** (Association for building communities and cooperatives) – can support start-ups.
- **Denmark's Samsø** – world-famous example of an island that became 100% renewable through cooperatives. A Swedish pilot can invite Samsø's experts as advisors.

What you can do tomorrow

If you are a farmer or landowner: Gather your neighbors. Join forces with the local community association. Invite a speaker from a Danish energy cooperative. See if there is wind or solar potential on your land. Contact the Swedish Energy Agency for a feasibility study grant.

If you are a municipal politician: Develop a policy that prioritizes cooperatives in land allocation for wind power and solar parks. Allocate land to cooperatives at a lower lease fee. Participate in subscribing shares in the cooperative as a municipality.

If you are a homeowner: Investigate whether your area has a solar cooperative. If not—start one. It does not need to be large-scale: 10 neighbors jointly buying a solar park of a few hectares is a cooperative.

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Peer-to-peer electricity trading

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Summary: Enables direct trading of solar electricity between neighbors (peer-to-peer) via a digital platform. A household with solar panels on the roof sells surplus electricity directly to the neighbor instead of to the grid owner for a low compensation. The platform matches buyers and sellers, handles payments, and reports to the grid owner. The technology already exists (blockchain, smart meters), but Swedish legislation (the Electricity Act) requires all electricity trading to go through a registered electricity trading company. The goal: to create fairer compensation for small-scale producers, increase incentives for solar cells, and let neighbors share both electricity and costs. Pilot in a city district or village with 500 households.

The mechanism

Today, a household with solar cells sells its surplus electricity to the grid owner (the local electricity grid company) at a low price—often 20–40 öre/kWh, far below the price the household itself pays for purchased electricity (100–300 öre/kWh). The grid owner then resells the same electricity to neighbors at market price. The difference is the grid owner's profit. The household with solar cells thus receives very poor payment while the grid owner profits from the solar cell's production.

The measure: Introduce **peer-to-peer trading** (P2P) where:

- A household (seller) with solar cells lists its surplus electricity on a digital platform.
- Another household (buyer) in the same local grid area (same low-voltage station) buys electricity directly from the seller via the platform.
- The platform matches bids and prices (e.g., the seller sets a price of 80 öre/kWh, the buyer accepts). The price lies between the grid owner's purchase price (low) and the regular electricity price (high)—both win.
- The electricity flows physically as usual via the grid, but the financial transaction goes directly between the neighbors. The platform reports to the grid owner so that the grid fee (which covers transport losses) can be billed separately.
- The technology is based on **smart electricity meters** (installed in all Swedish homes since 2020) and a digital platform (mobile app or web) with BankID login. Blockchain can be used to secure transactions, but is not necessary.

Example:

Anna has solar panels on her roof. On a sunny day, she produces 15 kWh, but uses only 5 kWh herself. 10 kWh normally go out onto the grid and she gets 25 öre/kWh from the grid owner = SEK 2.50. Her neighbor Bertil pays 150 öre/kWh for his electricity at the same time (including taxes and fees).

With peer-to-peer trading: Anna sells her 10 kWh directly to Bertil via the platform for 80 öre/kWh (agreed price). Anna gets SEK 8 (instead of SEK 2.50). Bertil pays SEK 8 for the electricity, plus a small grid fee to the grid owner (20 öre/kWh, total SEK 2)—a total of SEK 10, which is far less than the SEK 15 he would otherwise pay for the same amount of electricity (150 öre × 10 kWh). Both win. The grid owner still gets their grid fee and does not lose revenue.

Why this is not possible today:

- The Electricity Act requires that all electricity trading occurs via a registered electricity trading company that has balance responsibility (responsibility for matching production and consumption in real-time).
- Private individuals are not allowed to sell electricity directly to each other without such a middleman.
- Electricity grid companies have no incentive to enable peer-to-peer trading—they profit from the middleman difference.

The core of the measure: Amend the Electricity Act so that small-scale peer-to-peer trading (below a certain capacity limit, e.g., 100 kW per facility) is exempted from the requirement of an electricity trading company. The platform (cooperative or municipal) takes the role of "technical coordinator" but not as balance responsible—that responsibility remains with the grid owner.

Pilot design – 500 households in two areas

Pilot phase 1 (years 1-3): Two pilot areas with a total of 500 households test peer-to-peer trading under an exemption from the Electricity Act.

Pilot 1: Hammarby Sjöstad, Stockholm (urban area, mixed buildings)

- 300 households, of which 100 with solar panels (roof or balcony).
- The platform is developed by a local energy cooperative in collaboration with KTH.
- Grid owner: Vattenfall Eldistribution (which consents to the pilot).
- Goal: test the technology in an urban area with high solar cell pressure.

Pilot 2: Simris, Österlen (rural area, high solar cell density)

- 200 households, of which 80 with solar panels (many farms).
- The platform is developed by a community association in collaboration with Eon.
- Goal: test in an area with large overproduction during the day and low grid capacity.

Organizational model:

- **Platform development**: A foundation or cooperative (with the pilot areas as members) builds the app. Budget SEK 2–3 million (partially financed by the Swedish Energy Agency).
- **Technical solution**: The app connects to each household's smart electricity meter (API from the grid owner). Households enter buy/sell bids. The platform matches and sends payments via Swish or direct debit.
- **The grid owner's role**: The grid owner continues to measure and bill the grid fee (variable part) for all electricity passing through the grid. Peer-to-peer trading does not affect the grid owner's technical responsibility.
- **Legal exemption**: The government grants an exemption from the Electricity Act for the pilot. An electricity trading company is not required, but a "technical coordinator" (the platform owner) must report to the grid owner and the Swedish Energy Markets Inspectorate.

Technical specifications:

Component	Solution
Identification	BankID (seller and buyer)
Measurement data	API from the grid owner's measurement system (hourly values)
Matching	Auction algorithm (seller sets price, buyer accepts) or dynamic price (spot price + margin)
Payment	Swish (direct) or monthly invoice via the platform
Security	Encrypted communication, blockchain as optional (not a requirement)
Grid fee	The grid owner bills separately (variable fee per kWh, unchanged)

Evaluation:

- Number of active users after 6, 12, 24 months (target: >70% of households in the pilot)
- Average price of peer-to-peer traded electricity compared to the grid owner's purchase price and market price
- Sellers' extra revenue per year (compared to selling to the grid owner)
- Buyers' savings per year (compared to regular electricity price)
- Technical problems (downtime, measurement errors, payment disruptions)
- The grid owner's experience (does it affect grid stability? extra administration?)

Budget and financing

Item	Amount (2 pilots)
Platform development (app, backend, API integration)	SEK 2,500,000 (one-time)
Platform operation (server, support, 2 years)	SEK 500,000
Marketing and information to households	SEK 300,000
Legal advice (exemption, agreements)	SEK 200,000
Evaluation (independent researchers)	SEK 500,000
Total	**SEK 4,000,000**

Financing: Funds are applied for from the Swedish Energy Agency's program for "Smart grids and flexibility" (budget approx. SEK 200 million/year). SEK 4 million is a minor pilot. Once the technology is proven, the platform can be scaled up to other areas without additional state financing (the cost per household will be low, approx. SEK 500 per connection).

Regulatory changes required

1. **Amendment to the Electricity Act (1997:857)** – add a section: *"A household or small business may transfer electricity from a solar cell facility with an installed capacity below 100 kW directly to another end customer within the same grid area without being registered as an electricity trading company. The transfer shall take place via a digital platform that reports to the grid owner and the Swedish Energy Markets Inspectorate."*
2. **Ordinance on technical coordinators** – the Swedish Energy Markets Inspectorate is tasked with approving platforms for peer-to-peer trading. Requirements for security, transparency, and consumer protection.
3. **The Tax Ordinance** – peer-to-peer trading between private individuals is not subject to VAT (private individuals have no VAT registration). The electricity sold is already taxed via energy tax when fed into the grid? Clarification: peer-to-peer trading is a resale of already taxable electricity—no additional tax.
4. **General Data Protection Regulation (GDPR)** – households' electricity usage is sensitive personal data. The platform must have consent from the users and comply with GDPR. Simple templates for terms of use are produced.

Expected resistance and answers

"The grid owners lose revenue – they will block the pilot."

Answer: The grid owners still charge a grid fee for all electricity that passes through. Peer-to-peer trading does not reduce the physical amount of electricity in the grid (the electricity flows anyway). Some grid owners can even profit from peer-to-peer trading, as it reduces bottlenecks (when production and consumption occur locally, the overlying grid is not burdened). The pilots require the grid owner to consent and share measurement data—otherwise, the government can force them via the Electricity Act.

"Electricity trading companies lose customers – they lobby against it."

Answer: The electricity trading companies' business model is based on buying electricity cheaply (from producers) and selling expensively (to customers). Peer-to-peer trading competes with that margin. But electricity trading companies can adapt—for example, by offering peer-to-peer trading platforms themselves (as a service) or by focusing on larger customers. The pilot is small and poses no threat in the short term. In the long term, the electricity trading industry must change—it is inevitable with smart grids.

"The technology is not mature – blockchain is too slow and expensive."

Answer: Blockchain is optional, not necessary. Smart meters, cloud services, and encrypted APIs are sufficient. The platform can be a simple web server with a database. The cost per transaction will be a few öre—negligible. The technology is already proven in similar pilots in Germany, the Netherlands, and Australia.

"This is just for electricity nerds – ordinary people won't use it."

Answer: The platform is designed to be simple: the user enters an accepted price (e.g., "I want to buy for max 100 öre/kWh"). Everything else happens automatically. Many will appreciate the opportunity to get paid more for their solar electricity or to buy cheaper electricity from their neighbor. The pilot is intended to test user-friendliness precisely.

"Are there international examples?"

Answer: Yes. **Germany** (Pebbles project, SonnenCommunity)—thousands of households trade solar electricity peer-to-peer. **The Netherlands** (Vandebron, Powerpeers)—platforms with peer-to-peer trading. **Australia** (Power Ledger)—blockchain-based peer-to-peer trading in several city districts. **USA** (Brooklyn Microgrid)—local solar energy trading. Sweden is lagging behind—the pilot can provide Swedish experience.

Swedish examples to build upon

- **Own electricity usage with solar cells** – already today you can use your own solar electricity directly. The next step is to share with the neighbor.
- **Housing cooperatives' shared solar electricity** – many housing cooperatives (BRF) install solar panels on the roof and distribute the electricity among members (internally). This is a form of peer-to-peer trading within the same property. The scale model exists.
- **Local energy communities (EU Directive 2019/944)** – the EU's clean energy for all Europeans directive encourages member states to enable energy communities. Sweden has not fully implemented this. Peer-to-peer trading is a way to live up to the directive.
- **Elogen (Eon)** – a Swedish electricity trading company that offers hourly pricing, but not peer-to-peer trading. They could be a pilot partner.

What you can do tomorrow

If you have solar cells: Talk to your neighbors who also have solar cells. Ask the grid owner if they have plans to enable peer-to-peer trading. Write a joint letter to the Swedish Energy Markets Inspectorate.

If you are a grid owner: Get together with a few customers in a pilot project. You already have the measurement data and technical systems. Building a simple platform is not difficult. Be the first in Sweden—it is good PR.

If you are a politician (parliament or region): Submit a motion to introduce a legislative amendment for peer-to-peer trading (see the regulatory change proposal above). Use the EU Directive 2019/944 as an argument—Sweden has to implement it anyway.

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Energy banks

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Summary: Municipal or cooperative energy banks where households and companies can lend surplus capacity to the electricity grid during peak loads. An energy bank is a digital platform and aggregator that gathers small flexibility resources—home batteries, parked electric cars, heated water tanks, controlled heating (heat pumps, electric boilers), and sometimes smaller reserve power plants. When the electricity grid is overloaded (e.g., a cold winter evening in SE3), the grid owner pays the energy bank to reduce its capacity (load management) or feed in stored electricity. The revenue is distributed to the participating households. The goal: to reduce the need for expensive grid investments (new lines, transformers) and replace them with distributed flexibility—at a fraction of the cost.

The mechanism

Today, Svenska kraftnät and local grid owners expand the electricity grid to handle power peaks that occur only a few dozen hours per year (e.g., cold winter evenings when everyone is cooking, heating houses, and charging the car at the same time). A new 130 kV line can cost billions, but is fully utilized only 1–2% of the time. At the same time, hundreds of thousands of batteries, electric cars, and heat pumps sit unutilized—they could help shave the peaks.

The measure: An **energy bank** is a local aggregator that:

- Recruits households and companies that have flexible resources (see table below). The participant signs up via an app and receives a small bonus (e.g., SEK 500/year).
- Aggregates (gathers) thousands of small resources into a virtual power reserve—large enough for the grid owner to be able to buy the service.
- When the grid owner signals that there is a power shortage (price or direct command), the energy bank sends control signals to the participants' equipment: the battery is discharged, the electric car pauses charging, the water heater is turned off for 30 minutes, the heat pump lowers the output.
- The participant receives compensation based on how much power (kW) was freed up and for how long (hours). The compensation is often higher than the cost of the electricity not used—a net profit for the household.
- The energy bank can also store electricity when the price is low (nighttime, windy days) and sell it back when the price is high (peak shaving via batteries).

Examples of flexible resources in a standard household:

Resource	Typical power	Possible action	Impact on comfort	Compensation per hour (estimate)
EV charging	3–11 kW	Pause charging 1–2 h	None (the car is parked anyway)	SEK 10–30/h
Water heater	2–6 kW	Turn off 30–60 min	None (the water stays hot for a long time)	SEK 5–20/h
Heat pump (air/water)	1–5 kW	Lower output 30%	Maybe 1 degree colder for an hour	SEK 5–25/h
Home battery (5–15 kWh)	3–5 kW	Discharge during peak hour	None	SEK 20–50/h
Electric boiler (direct-acting)	3–10 kW	Turn off 15–30 min	The temperature drops marginally	SEK 10–40/h
Smart washer/dryer	1–3 kW	Delay start	None (the program runs later)	SEK 2–10/h

The societal benefit: By activating the energy bank during the 50–100 highest power peaks per year, the grid owner can avoid or postpone billion-kronor investments in grid reinforcement. Every kilowatt of peak load shaved saves SEK 5,000–15,000 in grid costs. The energy bank's cost is a fraction of that—typically SEK 500–1,000 per installed kW.

Pilot design – 3 local energy banks

Pilot phase 1 (years 1-4): Three energy banks in different types of geographies—an urban area, a medium-sized town, and a rural area. Each bank gathers at least 500 flexible resources.

Pilot 1: Stockholm (Hammarby Sjöstad / Nacka)

- Type: Urban area, high proportion of electric cars, batteries, and heat pumps.
- Goal: 1,000 households, total flexible capacity 2–3 MW.
- Grid owner: Vattenfall Eldistribution.
- Technology: Platform from a local energy cooperative + startup.

Pilot 2: Västerås (central town + surrounding towns)

- Type: Medium-sized town, mix of detached houses, tenant-owner apartments, and smaller industries.

- Goal: 800 households + 20 companies, flexible capacity 2 MW.
- Grid owner: Mälarenergi Elnät.
- Technology: Municipal platform (Mälarenergi develops its own).

Pilot 3: Skellefteå (sparsely populated area, rural area)

- Type: Sparsely populated area, many electric boilers, water heaters, and some electric cars.
- Goal: 500 households, flexible capacity 1.5 MW.
- Grid owner: Skellefteå Kraft Elnät.
- Technology: Cooperative platform (community association + research support).

Organizational model:

- The energy bank is run as an economic association (cooperative) or as part of the local energy office/municipal housing company.
- The participants sign an agreement that allows the energy bank to control their equipment (via API or manual opt-in). The agreement specifies the minimum compensation and the maximum number of control events per year (e.g., max 100 hours, so that comfort is not significantly affected).
- The grid owner buys the flexibility service from the energy bank via an agreement. The price is set through bidding or negotiation. Typically, the grid owner pays a fixed capacity fee (SEK per available kW per month) plus variable compensation (SEK per kWh freed up).
- The energy bank distributes the compensation to the participants (minus a small fee for operations, 10–20%).

Technical specifications:

Function	Solution
Communication with equipment	Local APIs (e.g., OCPP for charging boxes, Modbus for heat pumps) or via smart meters (controllable load)
Platform	Cloud-based, open source (enables more energy banks)
Signal from grid owner	Direct command via API (encrypted) or via price information (spot price)
Participant app	Simple app (iOS/Android) where the user sees their consumption, earned compensation, and can set comfort limits
Security	The equipment may only be controlled within agreed limits—no risk of unauthorized access

Evaluation:

- Total installed flexible capacity (kW) per energy bank
- Proportion of peak load that can be shaved (compared to a scenario without an energy bank)
- The grid owner's reduced investment needs (SEK million/year)
- Participants' satisfaction and the proportion who continue after the pilot
- Cost per flexible kW—compared to traditional grid reinforcement (the goal is at least 50% lower)

Budget and financing

Item	Amount (per pilot bank)	Total 3 banks
Platform development (open source, shared)	SEK 1.5 million (one-time)	SEK 1.5 million (shared)
Platform operation (3 years)	SEK 200,000	SEK 600,000
Marketing & recruitment (500–1,000 households)	SEK 300,000	SEK 900,000
Compensation to participants (for the pilot period)	SEK 500,000	SEK 1.5 million
Administrative resource (0.5 FTE × 3 years)	SEK 900,000	SEK 2.7 million
Evaluation (research)	SEK 500,000	SEK 1.5 million
Total	**SEK 3.9 million per bank (3 years)**	**~SEK 9 million** (shared costs reduce the total)

Financing: Funds can be applied for from the Swedish Energy Agency's programs for "Smart grids" and "Flexibility in the electricity system" (budget approx. SEK 200 million/year). EU funds (Horizon Europe, LIFE) can also be used. Once the pilot is evaluated, the business model can carry itself: the grid owner pays the energy bank for flexibility, and the energy bank pays the participants. The investment in the platform and start-up costs are one-time expenses.

Regulatory changes required

1. **The Electricity Act (1997:857)** – today, the grid owner is obliged to connect all customers but has no obligation to buy flexibility services. A new section should be introduced: *"The grid owner shall, in areas with limited grid capacity, strive to purchase flexibility services from aggregators (such as*

energy banks) before grid reinforcement is carried out, if it is socio-economically efficient." (Similar rules already exist in the UK and California.)

2. **Ordinance on grid fees** – the grid fee is differentiated so that customers who provide flexibility receive a lower fixed fee or a bonus (incentive). The Swedish Energy Markets Inspectorate is tasked with designing the model.
3. **Secrecy and GDPR** – the energy bank needs access to the participants' measurement data (kWh per hour). Requires consent according to GDPR. A template for consent is developed.
4. **Standard agreements** – The Swedish Energy Markets Inspectorate and the industry organization Swedenergy (Energiföretagen Sverige) develop a standard template for agreements between the grid owner and the energy bank, as well as between the energy bank and the participant. This lowers the threshold for new actors.

Expected resistance and answers

"The grid owners want to build grids – that is their core business. They will not buy flexibility."

Answer: The grid owner's mission is to deliver reliable electricity at the lowest possible cost for the customers. If flexibility is cheaper than grid reinforcement (which it often is), the grid owner is obliged under the Electricity Act to consider it. The pilot includes grid owners who are positive. To drive the transition, regulation is required (see above)—when the incentives are right, the grid owners will adapt.

"This is too complicated for ordinary households – who has the energy to configure their gadgets?"

Answer: The energy bank handles everything automatically after the user has signed up. No daily adjustments are required. The user just sets a comfort limit (e.g., "the car must be charged to 80% by 08:00" or "the temperature must not fall below 18 degrees"). The rest happens in the background. The app shows the earned compensation—which often motivates participation.

"Batteries and electric cars are expensive – this will only be for the rich."

Answer: Even households without a battery or electric car can participate via a water heater, heat pump, or washing machine. These resources are available in most homes. And the more who participate, the lower the costs for the grid—which benefits all customers (even those without flexible resources) through lower grid fees.

"Controlling equipment in the home is privacy-sensitive."

Answer: The participant decides for themselves when and how much the equipment may be controlled (e.g., max 2 hours per day, not between 17:00–19:00 if cooking). All data is encrypted. The agreement specifies that the energy bank may never turn off equipment that affects safety (e.g., frost protection). The privacy risk is low compared to the benefit that a lower electricity bill and reduced climate impact provide.

"Are there any Swedish examples?"

Answer: Yes, several small pilots. **E.ON** has tested load management of electric cars in the "Smart Charging" project. **Vattenfall** has tested virtual batteries ("V2G") in a few individual homes. **Göteborg Energi** has flexibility services for district heating. But no one has built an open, cooperative energy bank on a large scale. The pilot would be the first in Sweden—and could become a model for Europe.

Swedish examples to build upon

- **Load management of electric cars (E.ON, Vattenfall)** – already today, charging boxes can be controlled via an app. The technology exists.
- **Solar cell pools (cooperatives)** – use an aggregator to sell surplus electricity. The same model can be used for flexibility.
- **Grid benefit and flex market (Svenska kraftnät)** – they already have a market for large-scale flexibility (reserve power). The energy bank is a local, small-scale variant.
- **Denmark's "Energy Collective"** – a Danish energy cooperative that aggregates flexibility from homes. Sweden can visit and learn.

What you can do tomorrow

If you have an electric car, battery, or heat pump: Investigate whether your grid owner or your energy company offers control for compensation. Ask their customer service. The more people who ask for it, the faster it happens.

If you are a grid owner or energy office: Start a small pilot—gather 100 customers who are willing to control their charging or water heater. Offer a bonus of SEK 500/year. Measure the effect. Show that it works. Then build a platform.

If you are a politician (municipality or parliament): Submit a proposal to introduce a "flexibility obligation" for grid owners—that they must purchase flexibility services before building new grid capacity. Use the text above as a basis.

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Municipal energy councils

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Summary: Each municipality establishes an energy council consisting of residents, business representatives, local grid owners, environmental organizations, and municipal officials. The council has formal veto power (or decisive influence) over new large-scale energy projects—wind farms, large-scale solar parks, new power lines, hydrogen plants—planned within the municipality. The decision is based on local acceptance, impact on industries (reindeer husbandry, tourism, agriculture), and the environment. The goal: break the deadlock where central decisions are made without local anchoring, which leads to appeals, delays, and conflicts. By giving local actors real influence, predictability for project developers increases—they know that if the energy council says yes, there will be no appeals.

The mechanism

Today, large-scale energy projects are reviewed in several instances: the municipality (detailed development plan or building permit), the county administrative board (environmental permit), the land and environment court (appeals), and sometimes the government (permit under the Electricity Act). The municipality has a so-called "municipal veto"—it can say no to wind power within the municipal borders. But the veto is rarely used, because it is politically sensitive and can be overturned in some cases. The result: projects receive permits despite local protests, which leads to long appeal processes (5–10 years) and ultimately often that the project is abandoned.

The measure: A **municipal energy council** is a broadly composed group that:

- Is appointed by the municipal council for a term of office (4 years). The council has 11–15 members:
  - 3 residents (elected through citizen dialogue or lottery)
  - 2 representatives for local business (chamber of commerce, business association)
  - 1 representative for affected Sami villages (in northern Sweden) or farmers (in the south)
  - 1 representative for local environmental organizations (The Swedish Society for Nature Conservation, local initiatives)
  - 2 representatives for the local grid owner (Eon, Vattenfall, municipal electricity grids)
  - 2 politicians (from different parties, not the chair)
  - 2 officials (the municipality's head of environment and building permits, planning architect)
  - Chairperson (external, e.g., a retired judge or university teacher)

- The council has **veto power** (or a strong recommending role, with a requirement that the municipal council follows the decision unless 2/3 vote against). The veto applies to:
    - Wind farms with more than 3 turbines or a total height over 150 m.
    - Solar parks of more than 10 hectares.
    - New power lines of voltage class 130 kV or higher (which require a permit under the Electricity Act).
    - Hydrogen or battery parks above a certain size.
- The process:
    1. The project developer applies for an advance ruling from the energy council *before* formal permits are sought.
    2. The council requests documentation, holds public consultations, and gathers feedback.
    3. The council makes a decision within 6 months: "approved", "approved with conditions" (e.g., compensation measures to the local community), or "rejected".
    4. If the council rejects, the project developer cannot proceed (or must appeal to the land and environment court, but with a very high burden of proof). If the council approves, the project is in principle "pre-secured"—local appeals are not accepted in later instances (exception: national interests that cannot be resolved locally).

Why this solves today's problems:

Problem today	The energy council's solution
Municipal vetoes are rarely used (political cowardice)	The energy council is broad and apolitical – easier to say no
Projects are appealed by many different groups (villagers, environmental organizations, reindeer owners)	All groups are represented in the council – their views are handled early
The permitting process takes 5–10 years	The council's decision is made within 6 months – the rest of the process is faster
The project developer does not know if the project is accepted until after million-kronor investments	Advance ruling before large costs – reduces risk
Local protests lead to media attention and political crises	The council's decision is anchored – few surprises

Pilot design – 5 municipalities first

Pilot phase 1 (years 1-3): Five municipalities in different parts of the country establish energy councils with veto power (according to the model above). The government grants an exemption from the current Planning

and Building Act during the pilot period.

Proposed pilot municipalities:

Municipality	Why?	Expected conflict
Piteå	Planned large-scale wind power (Markbygden) – many protests from reindeer owners and villagers	High
Öland	Many plans for wind power and solar parks – the tourism industry is worried about the impact	High
Vara (Västra Götaland)	Agricultural area, resistance to solar parks on arable land	Medium
Åre	Wind power on the mountain – clash with tourism and reindeer husbandry	High
Halmstad	Offshore wind power (sea outside) – conflict between environmental interests and fishing	Medium

Organizational model:

- The council is appointed by the municipal council after an open application process (for residents, business, etc.).
- The council has a secretary (municipal employee or external consultant) who handles administration and investigations.
- The project developer pays a fee to have their case reviewed (SEK 50,000–200,000, depending on project size). The fee covers the council's costs.
- The council's decisions are public and must be justified in writing. The decision can be appealed to the land and environment court, but only by the project developer (if they consider the decision violates the law)—not by a third party (villagers, organizations). This is to reduce the number of appeals.

What happens if the council says no?:

- The project cannot be implemented within the municipality, unless the government or the land and environment court overturns the veto due to "compelling national interests" (e.g., national energy preparedness). This exception is used very restrictively.
- The project developer can propose compensation measures (e.g., build a new school, finance local road maintenance, provide annual compensation to the Sami village) and ask for a new review.

Evaluation:

- Number of projects approved, approved with conditions, and rejected.
- Average time from application to decision (target: <6 months).
- Proportion of the council's decisions that are appealed (target: <5%).
- Proportion of projects that are actually built after an approved decision (compared to today's 30–50%).
- Surveys to project developers, municipal politicians, and residents regarding satisfaction with the process.

Budget and financing

Item	Amount (per municipality and year)	Total 5 municipalities (3 years)
Secretary (0.5 FTE)	SEK 300,000/year	SEK 4.5 million
Members' remuneration (15 people × 10 meetings × SEK 2,000)	SEK 300,000/year	SEK 4.5 million
Investigations (external experts when needed)	SEK 200,000/year	SEK 3 million
Information and citizen dialogue	SEK 100,000/year	SEK 1.5 million
Evaluation	–	SEK 1 million (total)
Total	**SEK 900,000/year per municipality**	**SEK 14.5 million** (over 3 years)

Financing: The fee from project developers covers part of it (if 5 projects per municipality and year with a fee of SEK 100,000 = SEK 500,000/year). The remaining SEK 400,000/year per municipality can be financed by the municipality itself (as an investment in predictability and reduced conflicts) or by the state (via the Swedish Energy Agency). SEK 14.5 million over 3 years is a modest sum in context.

Regulatory changes required

1. **Amendment to the Planning and Building Act (2010:900)** – introduce a new section: *"The municipality may establish an energy council with veto power over large-scale energy projects according to the appendix. The council's decision shall be binding for the municipality and for subsequent permit review, unless the government decides otherwise for special reasons."*
2. **Amendment to the Environmental Code (1998:808)** – for projects approved by the energy council, the right of appeal by third parties shall be limited. This is to avoid local protests being moved from the municipality to the court. (Corresponding rules already exist for certain infrastructure projects.)

3. **Ordinance on the composition of energy councils** – the government establishes minimum requirements for representation (residents, business, Sami villages, environmental organizations). The municipalities can add more members.
4. **Appeal process** – the project developer's opportunity to appeal a rejection decision is reviewed by the land and environment court, which can only overturn the veto if it is "manifestly unreasonable" or violates "compelling national interests".

Expected resistance and answers

"The municipalities will shut everything down – no projects will happen."

Answer: On the contrary. Today, municipalities often say yes to projects that the local population opposes, which leads to appeals. With the energy council, the municipality gets a clear signal in advance—and can deny projects that would be stopped by appeals anyway. The developer avoids spending millions on projects that won't happen anyway. In Germany, where similar models exist (Bürgerenergiegenossenschaften), the proportion of built projects has increased.

"Environmental organizations and Sami villages will get too much power – they can block everything."

Answer: They have one vote each, not a majority. The council consists of many different interests. If all projects are blocked, the pressure from businesses and politicians will change the composition in the next term. The system is self-regulating.

"This is undemocratic – a small group will decide important energy issues."

Answer: The council complements, rather than replaces, the democratic process. The municipal council appoints the energy council and can alter its mandate. All decisions are public. Residents still have the opportunity to exert influence via municipal politics. The difference is that a broad group (with local expertise) makes decisions on technically complex issues—just like a building committee or environmental committee.

"Project developers will go straight to the government to bypass the veto."

Answer: The exception possibility exists ("compelling national interests") but should be used very restrictively—typically if the project is of crucial importance to the stability of the electricity system or national security. For most projects, the energy council's decision applies. Developers know this and adapt: they present better projects, listen to local opinions, and offer compensation.

"Are there any Swedish examples?"

Answer: Not exactly, but closely related. **Municipal wind power companies** (e.g., Skellefteå Kraft) are a form of local influence. **Consultation under the Environmental Code** is mandatory but rarely has decisive power. **Village communities and Sami villages' veto power** exist in some cases (Reindeer Husbandry Act), but not collectively. The pilot would be unique in Sweden but resembles models in **Germany** (Bürgerinitiativen with veto power in certain federal states) and **Scotland** (community benefit funds with veto). We can learn from them.

Swedish examples to build upon

- **Municipal wind power companies** – Skellefteå, Piteå, Övertorneå. They often have local reference groups, but no formal veto power.
- **Consultation under the Environmental Code (Chapter 6)** – statutory, but not binding.
- **Sami villages' right of consultation** – they have the right to express their views, but no veto.
- **Denmark's "Lokalplaner for vindmøller"** – municipalities in Denmark have detailed development plans that identify suitable areas. Residents can exert influence in the planning process. Our model goes one step further.

What you can do tomorrow

If you are a municipal politician: Propose in the municipal council to establish an energy council (in pilot form). Use the text above as a basis. Offer to participate in drafting the statutes.

If you are an engaged citizen (villager, Sami, environmentalist, entrepreneur): Gather like-minded people. Write a letter to the municipal executive board: "We want an energy council with real influence. Let us be part of designing it."

If you are a project developer (wind power, solar power, batteries): Support the idea—you win on predictability. Lobby industry organizations (Swedish Wind Energy Association, Swedenergy) to push for municipal energy councils. It is better than today's lottery.

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Local grid fees

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Summary: The electricity grid fee (the fixed and variable cost that every electricity customer pays to their local grid company) is determined locally by a user-cooperative board instead of by the grid monopolist. The model entails customers in a geographical area (e.g., a municipality or a city district) joining forces and forming a cooperative that takes over the management of the local low-voltage grid (the one on the street outside the house). The cooperative leases the grid from the large grid owner (Eon, Vattenfall, Ellevio) at cost price and is itself responsible for operation, maintenance, and fee setting. The goal: to lower grid fees by eliminating profit requirements and administrative overhead, increase local transparency and control, and provide incentives for energy efficiency and flexibility. The model is taken from "municipal utility districts" in the USA and cooperative grids in Germany (Schleswig-Holstein).

The mechanism

Today, electricity grid companies are natural monopolies – the customer cannot choose a grid owner. Each grid owner (Eon, Vattenfall Eldistribution, Ellevio, and about 40 smaller municipal companies) sets its own fees within the framework approved by the Swedish Energy Markets Inspectorate (Ei). The fees vary greatly: from SEK 0.50/kWh in some municipal grids to over SEK 1/kWh in metropolitan grids. The difference is primarily due to different profitability requirements (private grids have higher return requirements), administrative complexity, and investment needs. The customer has no influence – neither over the size of the fee nor over how the grid is managed.

The measure: A **cooperative grid model** where:

- A local cooperative (user cooperative) is formed by at least 500 electricity customers (households, companies, municipal operations) within the same low-voltage area (connected to the same transformer station).
- The cooperative negotiates with the large grid owner to **lease** the local grid (cables, transformers, service lines). The lease fee is based on the grid owner's actual capital cost (depreciation) and maintenance cost – no profit margin.
- The cooperative takes over operation, maintenance, metering, and billing. It can hire contractors (e.g., the same as before) or do parts itself.
- The cooperative sets the grid fee (fixed fee per month + variable fee per kWh) according to what is required to cover:

    - The lease fee to the large grid owner
    - Own operating costs (personnel, maintenance, administration)
    - Provision for future investments (new cables, transformers)
    - Any surplus (refunded to members or invested in energy efficiency)
- The cooperative is governed by a board elected by the members (one member, one vote). All decisions regarding fees, investments, and operations are public.

Difference compared to today's model:

Aspect	Today's grid company (Eon, Vattenfall etc.)	Cooperative grid
Owner	Private or municipal limited companies	The members (the customers)
Profit requirement	Yes (often 6–8% return on capital)	No (non-profit-maximizing)
Administrative overhead	High (group management, marketing, lawyers)	Low (only local operation)
Fee setting	Determined by Ei (based on the company's proposal)	Determined by the members at the annual meeting
Transparency	Low (annual report public, but complex)	High (all board decisions, members can attend)
Flexibility	Low (standardized fees, few exceptions)	High (differentiation between customers, discount for flexibility)

Potential savings: A cooperative grid fee can be 20–40% lower than today's private grids, as profit requirements and corporate overhead disappear. For an average homeowner (approx. 20,000 kWh/year), this means a SEK 1,000–2,000 lower grid cost per year.

Pilot design – 3 cooperative grids first

Pilot phase 1 (years 2-5): Three areas (rural, suburb, smaller town) form cooperatives and lease the local grid from the large grid owner. The government enables the pilot via an exemption from the Electricity Act.

Proposed pilots:

Location	Type	Number of customers	Current grid owner	Expected savings
Hammarby Sjöstad (Stockholm)	Urban area, mixed	2,000 households + companies	Ellevio	15–20%
Östergötland (rural, Linköping municipality)	Sparsely populated area, farms, detached houses	800 households	Eon	25–30%
Kiruna (central town + surrounding areas)	Smaller town, mining community	3,000 households	Vattenfall Eldistribution	20–25%

Steps to form a cooperative grid:

1. **Interest group**: At least 10% of the customers in the area sign a letter of intent. The municipality or an existing association (e.g., community association) takes the initiative.
2. **Feasibility study**: The cooperative receives a grant (SEK 200,000 from the Swedish Energy Agency) to investigate the grid's technical condition, cost structure, and potential savings.
3. **Negotiation with the grid owner**: The cooperative negotiates a lease agreement specifying the lease fee (based on the grid owner's reported capital costs), division of responsibilities (who fixes faults, who invests in new cables), and transfer of metering data.
4. **Formation of the cooperative**: An economic association is registered with the Swedish Companies Registration Office (Bolagsverket). The members subscribe for shares (typically SEK 1,000–5,000 per household). The shares are used as working capital.
5. **Takeover**: The cooperative signs agreements with contractors for operation and maintenance (same as before or new ones). Meter reading and billing are set up.
6. **Operations**: The cooperative bills the members for the grid fee (lower than before). The surplus is refunded or invested in smarter grids (batteries, load management).

Risk management:

- The cooperative takes out insurance for unforeseen breakdowns (e.g., a transformer fire).
- A reserve fund is built up (10% of the annual fee) for future reinvestments.
- The lease agreement contains a "repossession clause": if the cooperative mismanages (e.g., goes bankrupt), the grid owner can repossess the grid at no extra cost.

Evaluation:

- Reduction of the grid fee in percentage compared to before (target: >20%).
- Members' satisfaction (survey).
- Operating cost per customer (compared to the grid owner's historical costs).
- Number of operational disruptions (should be unchanged or lower).

- Proportion of members actively participating in annual meetings and board elections (target: >15%).

Budget and financing

Item	Amount (per pilot)	Total 3 pilots
Feasibility study grant (SEK 200,000)	SEK 0.2 million	SEK 0.6 million
Legal assistance (agreements, registration)	SEK 100,000	SEK 0.3 million
Technical consultant (grid mapping)	SEK 150,000	SEK 0.45 million
Start-up capital (member contributions, covers working capital)	SEK 1–2 million (loans/contributions)	SEK 3–6 million (not a state cost)
Administrative support (first year)	SEK 300,000	SEK 0.9 million
Evaluation	SEK 200,000	SEK 0.6 million
Total state cost	**SEK 0.95 million per pilot**	**SEK 2.85 million**

Financing: The state's cost (SEK 2.85 million) is very low in relation to the long-term savings for electricity customers. EU funds (LIFE, Horizon) could possibly also be used. The start-up capital from members is an investment that yields a return in the form of lower grid fees (as a rule, the contribution is repaid within 2–4 years). The municipality can act as a guarantor or co-financier.

Regulatory changes required

1. **Amendment to the Electricity Act (1997:857)** – a new section enabling user cooperatives to take over the management of the local grid: *"The electricity grid concession holder (the grid owner) is obliged, at the request of a cooperative consisting of at least 500 customers within a defined grid area, to grant the right of use to the grids within the area for a reasonable fee corresponding to the grid owner's capital and operating costs."*
2. **Amendment to the Economic Associations Act (1987:667)** – clarification that cooperatives for grid management are permitted and that profit distribution (refunds) to members is tax-free if it is based on surplus from the fees.
3. **The Swedish Energy Markets Inspectorate's regulations** – Ei must approve the lease fees (so that the grid owner does not charge unreasonably high fees). A model for cost-based lease fees is developed.
4. **The Swedish Tax Agency** – clarify that the lower grid fee is not a taxable benefit for the members, as it is a refund of surplus from a cooperative association.

Expected resistance and answers

"The grid owners (Eon, Vattenfall, Ellevio) will oppose this by all means – it is their most profitable business."

Answer: Yes, they will lobby against it. But the government can force them via the Electricity Act. The grid owner still gets their capital cost covered via the lease fee – they only lose the extra profit that exceeds a reasonable return. It is difficult for the grid owner to argue that this profit is socially beneficial. In Germany and the USA, similar models have been introduced despite strong opposition from energy companies.

"The cooperatives lack competence – who will fix the grid when there is a fault?"

Answer: The cooperative hires the same contractors as the grid owner uses (or others). The difference is that the cooperative itself decides who gets the assignment and at what price. Many smaller electrical companies are interested in taking over maintenance from the large companies. The competence exists in the market.

"It will be expensive for the cooperative to build up administration and billing."

Answer: Billing can be outsourced to a smaller billing company or managed via a digital platform (costing a few kronor per customer and month). Administration can be minimal – a part-time coordinator is sufficient for 500–1,000 customers. The large grid companies have heavy administration with management teams, marketing, and lawyers – the cooperatives avoid this.

"What happens when the grid needs to be reinforced? The cooperative might not have capital."

Answer: The lease agreement regulates the investment responsibility. Normally, the grid owner owns the cables and transformers – the cooperative is responsible for operation and maintenance, but major reinvestments (e.g., a new transformer station) are made by the grid owner (and financed via a raised lease fee). The cooperative can also choose to finance certain investments itself if it is cheaper. There is flexibility.

"Are there Swedish examples?"

Answer: No full-scale ones, but closely related ones. **Municipal electricity grids** (e.g., Malmö, Gothenburg) are a form of local ownership, but they are still monopolies with political governance, not member-controlled cooperatives. **The Swedish Association of Energy Societies** (Sveriges Energiföreningars Riksförbund, SER) has pushed the issue of local grid cooperatives for many years. **Fiber cooperatives** are widespread in

Sweden (thousands of villages have built their own fiber networks). The technology and organizational model are proven – why shouldn't electricity grids be able to work in the same way?

Swedish examples to build upon

- **Fiber cooperatives** (e.g., Bygdenät, local fiber associations) – over 1,000 villages and city districts in Sweden have their own fiber networks. The members own the network and pay low fees. The same model for electricity grids.
- **Municipal electricity grids** (e.g., Malmö Energi, Göteborg Energi) – often have lower fees than private grids. Shows that non-profit-maximizing ownership lowers costs.
- **Cooperative wind farms** (Denmark, Germany) – member-owned and governed. Experience with member democracy and financing.
- **Norway's "lokale nettselskap" (local grid companies)** – Norway has many small grid companies, often owned by municipalities or local energy agencies. Some of them have cooperative elements.

What you can do tomorrow

If you are the chair of a housing cooperative or a community association: Talk to the neighbors. Check which electricity grid company you have and what you pay. Request a quote from the grid owner to lease the grid. Even if they say no – you have started a process.

If you are a municipal politician: Submit a proposal that the municipality should investigate the possibility of forming a municipal grid cooperative (in collaboration with municipal housing companies). Use fiber cooperatives as a model.

If you are a Member of Parliament: Submit a motion for a legislative amendment that forces grid owners to grant the use of grids to cooperatives (see the proposal above). Highlight the thousands of Swedish fiber cooperatives as proof that the model works.

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